



CIVITAS indicators

Share of deliveries to parcel lockers and drop-off points (TRA_FR_LK1)

DOMAIN

 Transport	 Environment	 Energy	 Society	 Economy
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TOPIC

Freight

IMPACT

Parcel lockers and drop-off points for freight urban deliveries

Increasing the use of parcel lockers and drop-off points for urban logistics

TRA_FR

Category

Key indicator	Supplementary indicator	State indicator
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CONTEXT AND RELEVANCE

Motorised freight transport refers to the movement of goods using motor vehicles such as trucks and vans. This mode of transport is widely used to deliver goods to customers in urban areas, but it contributes significantly to energy consumption, emissions, noise, and space occupancy. These factors negatively impact environmental sustainability and quality of life in cities.

Alternative solutions for urban deliveries, such as cargo bikes, parcel lockers and drop-off points, autonomous bots, drones, and shared logistics, can help reduce the reliance on conventional motorized freight transport. These alternatives contribute to lower emissions, reduced noise pollution, and improved space efficiency in urban areas.

This indicator measures the share of deliveries made to parcel lockers and drop-off points in the experiment area. Parcel lockers are self-service cabinets where recipients can pick up or drop off parcels, while drop-off points are staffed locations, such as shops, offering the same service. Both solutions increase convenience for recipients and reduce delivery distances, as vans do not need to visit the address of each recipient. They also eliminate failed home deliveries further improving the efficiency of the logistics operations.

This indicator is relevant when the policy action aims to increase adoption of parcel lockers and drop-off points by logistics providers for delivering goods in the pilot area. A successful action is reflected in a HIGHER value of the indicator.

DESCRIPTION

The indicator is the **share of deliveries made to parcel lockers and drop-off points in the experiment area**.

The unit of measurement of the indicator is %.

METHOD OF CALCULATION AND INPUTS

The indicator should be computed exogenously, by applying the method described and then coded in the supporting tool.

Method 1		
Calculation based on data from logistics operators	Significance: 0.75	
INPUTS		
The following information is needed to compute the indicator:		
a) The number of deliveries made to parcel lockers and drop-off points by logistics providers in the experiment area over a reference period		
b) The total number of deliveries made by logistics providers in the experiment area over the same reference period		
The experiment would result in a modification of the number of deliveries to parcel lockers and drop-off points in the experiment area.		

METHOD OF CALCULATION

The indicator should be computed **exogenously** according to the following steps:

- **Retrieval of the number of deliveries to parcel lockers and drop-off points and retrieval of the total number of deliveries in the experiment area.** If in the 'before' scenario no parcel lockers and drop-off points are used for deliveries in the experiment area, this value equals zero. The data should be obtained from the logistics operators active in the pilot area and should cover a relevant time period, such as an 'average day' reflecting normal operations.
- **Calculation of the index** by computing the ratio between the number of deliveries to parcel lockers and the total number of deliveries in the pilot area.

EQUATIONS

The equation computing the index is the following:

$$LKIndex = \frac{LKDel}{TotDel} \times 100$$

Where:

LKDel = Number of deliveries to parcel lockers and drop-off points in the experiment area over a reference period

TotDel = Total number of deliveries in the experiment area over the same reference period

ALTERNATIVE INDICATORS

This indicator measures the share of deliveries to parcel lockers and drop-off points in the experiment area. Indicator **TRA_FR_LK2** measures instead the average utilisation rate of parcel lockers, i.e., how much they are used relative to their capacity. The CIVITAS Evaluation Framework also includes indicators to assess the uptake of other alternative urban freight delivery modes: **TRA_FR_CB** for cargo bikes, **TRA_FR_AB** for autonomous delivery bots, **TRA_FR_PT** for parcels delivered via public transport, and **TRA_FR_DR** for urban air logistics.